AgriGuard

Working Prototype Walkthrough

All results below are from an actual live test run of the system, not mockups.

Live Working Dashboard (click to open):
https://handwoven-seduce-degree.ngrok-free.dev/dashboard

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1. AI Disease Detection Pipeline

A trained CNN (PlantVillage dataset, ~92% validation accuracy) classifies 38 disease categories across 14 crop species. Images arrive via a POST to /predict from any camera source (video pipeline, phone camera, or live streaming capture), are automatically corrected for any resolution or aspect ratio without distortion, and classified in real time.

- Example real result: Row 1 - Tree 1 -> Strawberry Leaf Scorch, 67% confidence -> treatment: fungicide (captan), remove infected leaves after harvest.
- Each result also returns an adaptive recommendation: quick_pass (healthy), requeue (low-confidence, re-check later), or extra_orbit (high-confidence, needs closer inspection now).

2. Live Farmer Dashboard (/dashboard)

A bilingual (English/Tamil) live dashboard, auto-refreshing every 5 seconds, built for a non-technical farmer to read at a glance.

- Hero status banner: overall field health (good / needs attention / urgent), with the count of trees currently flagged.
- Mission status bar: drone connection state, current mission phase, battery (simulated), and total trees scanned - all computed live from real detection activity.
- Action Needed panel: one card per diseased position, showing the disease, treatment, visit history/trend (worsening / improving / new), and the drone's recommended next pass.
- From tonight's test run: 13 positions inspected, 13 flagged for action, 0 healthy - an honest reflection of the test footage used, not an error.

3. Autonomous Flight Mission Plan (/mission-plan)

A 2D top-down simulation of the drone's planned survey - clearly labeled 'flight path simulated, hardware phase 2' for full transparency. Running the built-in simulation animates a drone icon through three distinct phases:

- Phase 1 - Grid Survey: visits every tree in a 5x3 field grid, showing the REAL detection result captured at each position as it goes.
- Phase 2 - 360-degree Close Inspection: automatically revisits only the positions the adaptive policy flagged for extra_orbit, with a real treatment recommendation shown per tree.
- Phase 3 - Low-Altitude Soil Pass: checks soil moisture and irrigation status before returning to base, ending with a real, freshly generated PDF report link.

4. Reports and Voice Summaries

Every inspection cycle can generate a bilingual PDF report (summary stats, disease breakdown, per-position detail, full activity log) and a spoken-language voice summary (via text-to-speech) in English or Tamil, for farmers who cannot read.

- Example real voice summary (English): "Field health update. 13 plants inspected today. 13 positions currently show disease. The most affected is Row 1 - Tree 1. Condition: Strawberry Leaf scorch. Recommended treatment: Apply fungicide (captan); remove infected leaves after harvest."

5. Live Camera Capture and Detection

A mobile-optimized camera page (/capture) lets any phone act as the input camera - streaming frames directly to the detection pipeline while a second screen (/live) shows the captured frame and AI verdict updating in real time, side by side.

What is real vs. concept

Disease detection, the adaptive flight-decision policy, the live dashboard, PDF/voice reports, and camera-based live capture all run on real data today. The physical drone airframe itself is the next hardware phase - the software above is built to work identically once it is airborne.